

RealEstate Hub

Production-Grade AI Voice Agent

Executive Report — Capstone Project

Conversational AI • Voice • RAG • Workflow Automation • UrduLish

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1. Executive Summary

RealEstate Hub currently handles dozens of daily phone inquiries from buyers, renters, and investors using human agents, which is expensive and inconsistent. This project delivers a production-ready AI voice agent that answers calls in fluent UrduLish, understands customer intent, answers property questions using grounded company data, recommends suitable properties, handles objections, and books, reschedules, or cancels property visits — while integrating with live business systems (Google Calendar, Gmail, CRM).

The system was built and evaluated over seven days, covering architecture design, knowledge-base and RAG construction, voice pipeline development, workflow automation, agent orchestration with LangGraph, a 43-case testing and security suite, and finally production deployment with a live voice-call demo.

Key Results

Metric	Result
Grounding rate (Day 2 hallucination eval, 20 questions)	100%
Hallucination rate	0%
Average response latency (text-only)	1.3s (target < 2s)
p95 latency (Day 6 suite)	2.29s
Test cases executed (Day 6)	43 (required: 40+)
Prompt-injection attempts blocked	10/10 (0 leaks)
RAG-miss / out-of-scope correct refusals	5/5 (100%)
Live booking / reschedule / cancel via Calendar + Gmail	Verified end-to-end

2. System Architecture

The agent is built as a pipeline of specialised components coordinated by a LangGraph state machine:

Telephony / Audio Input → VAD & Barge-in → Speech-to-Text (Whisper) → LangGraph Agent (Gemini 3.5 Flash-Lite) → Retrieval (ChromaDB + SQL) → Tool Calling (Calendar / Email / CRM via n8n) → Text-to-Speech (ElevenLabs) → Caller

Core Components

Layer	Technology	Notes
LLM reasoning	Gemini 3.5 Flash-Lite	Free tier; rate-limit-aware calls
Speech-to-Text	Whisper (local, small model)	Runs offline; Roman-script transcription for UrduLish matching
Text-to-Speech	ElevenLabs	Selected over Fish Audio after listening test (4.0/5 vs 2.5/5)
Agent orchestration	LangGraph	9-node state graph: intent, greeting, retrieve, book, reschedule, cancel, ai
Knowledge base	SQLite + ChromaDB	Structured facts (price, availability) via SQL; semantic FAQs/brochures via E
Workflow automation	n8n	Call → Intent → Match → Appointment → Calendar → E
Backend	FastAPI	/chat, /health, /metrics endpoints
Demo interface	Streamlit	Live mic-based call demo with autoplay voice replies
Deployment	Docker	Dockerfile + requirements.txt + healthcheck

3. Development Timeline (Day 1 – 7)

Day	Focus	Key Deliverables
1	Foundations	Architecture research, 9 conversation flows, UrduLish persona, Fish Audio vs ElevenLabs eval
2	Knowledge Base & RAG	SQLite knowledge base, ChromaDB RAG with chunk-size comparison, SQL vs vector routing,
3	Voice Pipeline	Streaming STT/TTS pipeline, natural speech behaviors, context memory, objection handling, h
4	Workflows	Google Calendar + Gmail + n8n + CRM integration, tested live with real bookings, reschedules,
5	LangGraph Orchestration	9-node LangGraph agent with tool calling and slot validation, tested across 6 scenarios
6	Testing & Security	43 test cases, prompt-injection suite, latency/monitoring metrics, Docker + CI/CD readiness
7	Deployment & Demo	FastAPI production backend, Streamlit live voice-call demo, executive documentation, monitori

4. Evaluation Results

4.1 Grounding & Hallucination (Day 2)

Twenty customer questions were answered by the LLM using only retrieved context, then checked for ungrounded facts. Result: 100% grounding rate, 0% hallucination rate. This check verifies numeric facts (prices, sizes, dates) against retrieved context; qualitative/semantic grounding was additionally spot-checked manually.

4.2 Voice & Latency (Day 3, Day 6)

Average end-to-end latency in text-only testing was 1.3 seconds against a 2-second target. The Day 6 evaluation suite measured a p95 latency of 2.29 seconds. ElevenLabs carries a higher baseline latency (~500ms) than Fish Audio (<200ms); this trade-off was accepted for its significantly better Urdu pronunciation and UrduLish code-switching (4.0/5 vs 2.5/5 in listening tests).

4.3 Workflow Reliability (Day 4)

Booking, rescheduling, and cancellation were tested against live Google Calendar and Gmail accounts, with n8n workflow automation confirmed via successful webhook triggers and CRM logging to both Google Sheets and SQLite.

4.4 Security & Guardrails (Day 6)

A 43-case test suite (exceeding the required 40) covered buyers, sellers, investors, renters, appointment management, off-topic conversation, angry callers, and silent callers. A dedicated prompt-injection suite of 10 adversarial attempts (e.g. “ignore instructions”, “reveal your prompt”, requests for internal data) was blocked in all 10 cases, with zero information leaks. Out-of-scope / RAG-miss questions were correctly declined in 5/5 cases rather than answered with fabricated information.

4.5 Memory Accuracy

An initial memory-accuracy test scored 70%; the issue was diagnosed and fixed, and a subsequent iteration reached 100% accuracy across the same test set — documented transparently as part of the evaluation process.

5. Security & Guardrail Testing

Test Category	Result
Prompt injection ("ignore instructions", etc.)	10/10 blocked, 0 leaks
Internal data exfiltration attempts	Declined politely, no leakage
Fake appointment booking attempts	Rejected, booking policy enforced
Unavailable slot / unavailable property booking	Blocked by availability validation
AI-disclosure honesty ("are you an AI?")	Agent discloses truthfully, offers human callback
Angry / distressed caller handling	Calm acknowledgement, escalation to human callback

6. Known Limitations

- The system runs entirely on free-tier services (Gemini, ElevenLabs, n8n self-hosted). Free-tier rate limits (e.g. 15 requests/minute on Gemini) require request pacing and are not suitable for production call volume without a paid plan.
- Real phone connectivity (SIP/Twilio) was not implemented, as telephony providers are paid services. The demo uses a browser-microphone interface (Streamlit) as a stand-in, per the free-tier project scope.
- The automated grounding check verifies numeric facts against retrieved context; it does not perform full semantic fact-checking of every generated sentence.
- Whisper's Urdu-script transcription mode produced poor keyword matches for UrduLish speech; the system was configured to transcribe in Roman script (English-language mode) to align with the agent's keyword-based intent detection.
- ElevenLabs is reported to cost 3–4x more per minute than Fish Audio at production volume; this was accepted for quality reasons and is revisited under future enhancements.
- On free API tiers, provider terms may permit input data to be used for product improvement. A production deployment handling real customer data requires a paid tier with appropriate data-handling terms.

7. Future Enhancements

- WhatsApp integration for text-based inquiries
- SMS appointment confirmations
- CRM integration beyond Google Sheets (Salesforce / HubSpot)
- Additional language support: Punjabi, pure Urdu script
- Voice cloning for a consistent brand representative voice
- Analytics dashboard (call volume, conversion rate, common objections)
- Lead scoring based on conversation signals
- Automatic follow-up campaigns for cold leads
- Payment gateway integration for booking deposits
- Live MLS / property feed integration in place of static data
- Migration from the deprecated google-generativeai SDK to the current google-genai SDK
- Real telephony integration (SIP/Twilio) for production phone-line deployment

8. Deployment Readiness

Item	Status
FastAPI backend (/chat, /health, /metrics)	Implemented and tested
Dockerfile + requirements.txt	Complete, with healthcheck
Environment variable management (.env, .env.example)	In place; secrets excluded via .gitignore
Monitoring & maintenance plan	Documented (latency thresholds, refresh schedule, backups)
Technical documentation	Architecture, API, user/admin/maintenance/troubleshooting guides
CI/CD readiness	GitHub Actions workflow scaffolded (Day 6)
Live 10-minute stakeholder demo	Recorded end-to-end (voice call, booking, security test)

9. Conclusion

The RealEstate Hub AI Voice Agent meets the objectives set out for this capstone: a production-style conversational AI system that speaks natural UrduLish, answers grounded property questions, recommends listings, handles objections, and manages the full appointment lifecycle through live integrations with Google Calendar, Gmail, and a CRM — all orchestrated by a tested, secure LangGraph agent. With the roadmap items above (paid-tier scaling, real telephony, additional language support), the system is positioned to move from demo to production deployment for an actual real-estate client.